ScientoPy v 3.0.1, User Manual

Juan Pablo Ruiz Rosero
jpabloruiz@unicauca.edu.co

Contents

1 Installation **3**

1.1 Pre-built releases (recommended) . . . 3

1.2 From source . . . 3

2 Interface overview **3**

2.1 Menu bar . . . 4

2.2 Keyboard shortcuts . . . 4

2.3 Status bar . . . 4

2.4 Configuration persistence . . . 4

3 Download the bibliometric dataset **5**

3.1 Download the dataset from Scopus . . . 5

3.2 Download the dataset from WoS . . . 5

4 Preprocessing (Tab 1) **6**

5 Analysis (Tab 2) **7**

5.1 Analysis parameters (left panel) . . . 8

5.2 Custom topics (right panel) . . . 8

5.3 Action buttons (bottom bar) . . . 8
5.4 Criterion descriptions . . . 9
5.5 Extract top topics . . . 9
5.6 Analyze custom topics inside a criterion . . . 9
5.7 Asterisk (*) wildcard . . . 10
5.8 Evolution plot . . . 10
5.9 Finding trending topics . . . 10
5.10 Analysis based on the previous results . . . 11

6 Results tab (Tab 3) 11
6.1 Column tooltips . . . 12
6.2 Action buttons . . . 12

7 Extended Results tab (Tab 4) 12
7.1 Action buttons . . . 13

8 Export tab (Tab 5) 13

9 Generate BibTeX 14

10 Output files and directories 14

11 ScientoPy graph types 16
11.1 Time line graph . . . 16
11.2 Horizontal bars graph . . . 17
11.3 Horizontal bars trends . . . 17
11.4 Evolution graph . . . 18
11.5 Word cloud graph . . . 18

1 Installation

1.1 Pre-built releases (recommended)

1. Download the latest stable release for your platform from: `https://github.com/jpruiz84/ScientoPy/releases`

- **Windows:** Download `ScientoPy-windows-x64.zip`, unzip and run `ScientoPyGui.exe`
- **Linux:** Download `ScientoPy-linux-x64.tar.gz`, extract and run `./ScientoPyGui`
- **macOS:** Download `ScientoPy-macos-arm64.zip`, unzip and run `./ScientoPyGui`

Note (Windows): On first run, Windows SmartScreen may show a warning. Click "More info" then "Run anyway". This occurs because the application is not code-signed.

1.2 From source

1. Clone the repository: `git clone https://github.com/jpruiz84/ScientoPy.git`

2. Create a virtual environment and install dependencies:

```
cd ScientoPy
python -m venv venv
source venv/bin/activate   # Linux/macOS
venv\Scripts\activate      # Windows
pip install -r requirements.txt
```

3. Run the GUI: `python ScientoPyGui.py`

2 Interface overview

ScientoPy uses a PySide6 (Qt 6) graphical interface with five tabs, a menu bar, and a status bar. The window is resizable with a minimum size of 853×480 pixels. The five tabs are:

1. **1. Pre-processing** – select the dataset folder and run the parallel Polars-based preprocessing pipeline.
2. **2. Analysis** – configure and run a scientometric analysis.
3. **3. Results** – browse the top-topic table from the last analysis (or the preprocessing brief right after a preprocess run).
4. **4. Extended Results** – per-document table with the papers that contributed to each top topic.
5. **5. Export** – on-demand CSV export of the preprocessed corpus or the analysis results (Scopus- or WoS-style fields).

2.1 Menu bar

- **View** → **Appearance:** Switch between Light, Dark, and System theme modes. The selected mode is saved and restored on next launch.

2.2 Keyboard shortcuts

- **Ctrl+O** – Browse and select a dataset folder
- **Ctrl+R** – Run (preprocess on Tab 1, analysis on Tab 2)
- **Ctrl+1/2/3/4/5** – Switch to the corresponding tab
- **Ctrl+C** – Copy selected cells in the Results or Extended Results tables

2.3 Status bar

The status bar at the bottom of the window shows feedback messages such as "Preprocessing complete – 5,548 papers", "Analysis complete – 10 topics found", or error information.

2.4 Configuration persistence

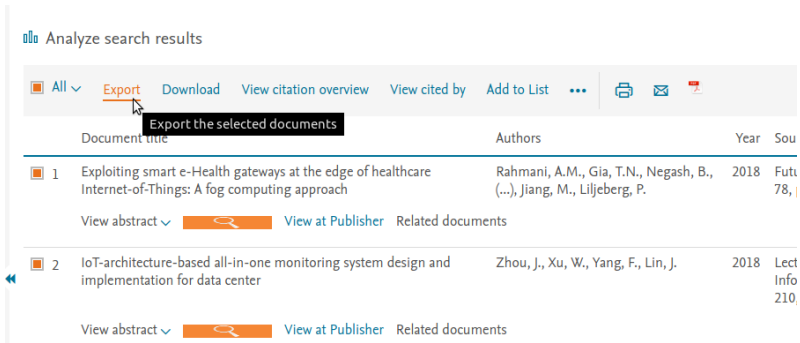
ScientoPy saves the following settings to `scientopy_config.json` and restores them on next launch: window size and position, splitter proportions, appearance mode, last dataset folder, and all analysis parameters (criterion, graph type, year range, topics length, skip first, window width, and checkbox states).

3 Download the bibliometric dataset

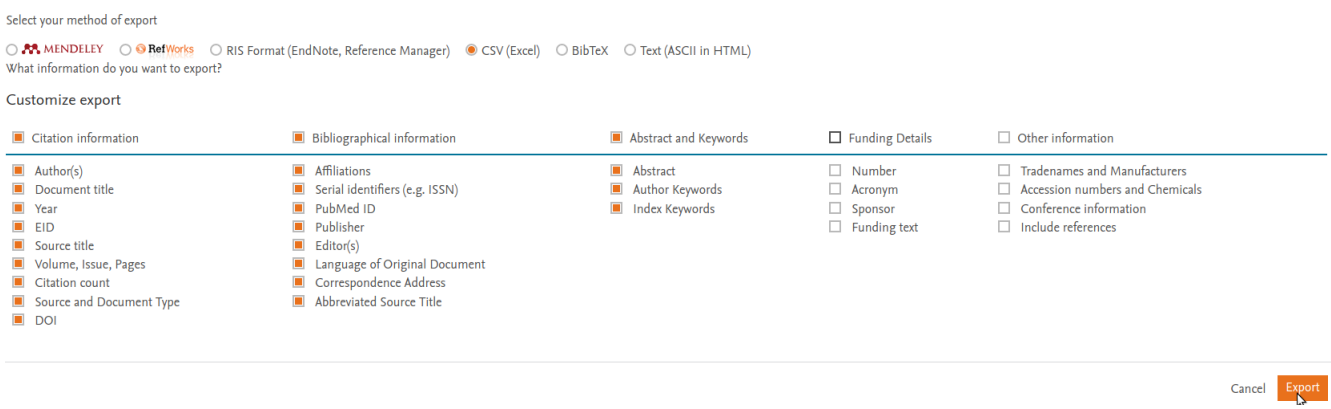
This section describes how to download the proper dataset from Scopus and WoS. Define a search criteria that will be used for Scopus and WoS. For this guide and for the example dataset we are using: "Bluetooth low energy".

3.1 Download the dataset from Scopus

1. Make your search with the defined search criteria for Article title, Abstract, Keywords.
2. Select all the results and click on Export:



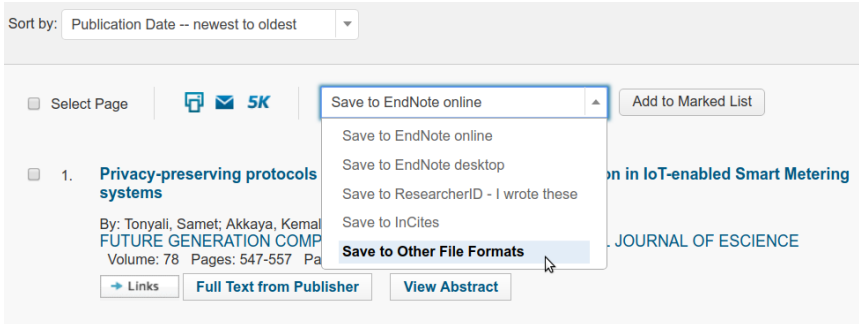
3. Select as method of export **CSV (Excel)**, and select the Customize export **Citation information, Bibliographical information, Abstract and Keywords**, then click on Export:



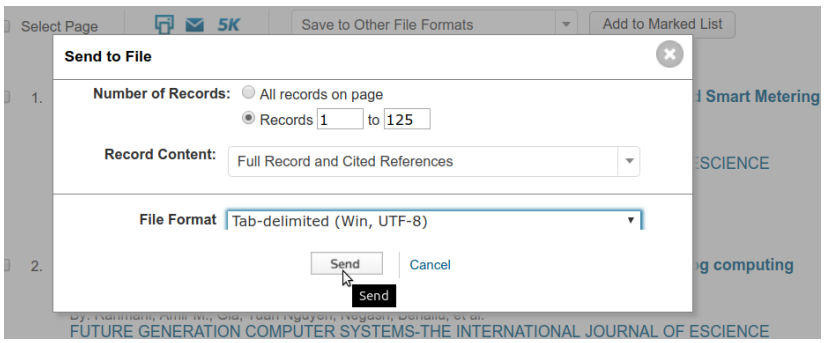
4. Save the file on the dataset folder (e.g. `/ScientoPy/dataIn`)

3.2 Download the dataset from WoS

1. Make your search with the defined search criteria for Topic.
2. Select **Save in Other File Formats**



3. Select the number of records to download, on Record Content select **Full Record and Cited References**, on File Format select **Tab-delimited (Win, UTF-8)**, and click on Send.



4. Save the file on the dataset folder (e.g. `/ScientoPy/dataIn`)

4 Preprocessing (Tab 1)

First we need to preprocess the downloaded dataset. This preprocess merges all the downloaded files from the dataset folder into a single file. Also, this process removes the duplicated documents. To preprocess the example dataset ("Bluetooth low energy" located in `dataInExample`), open ScientoPy and follow the next steps:

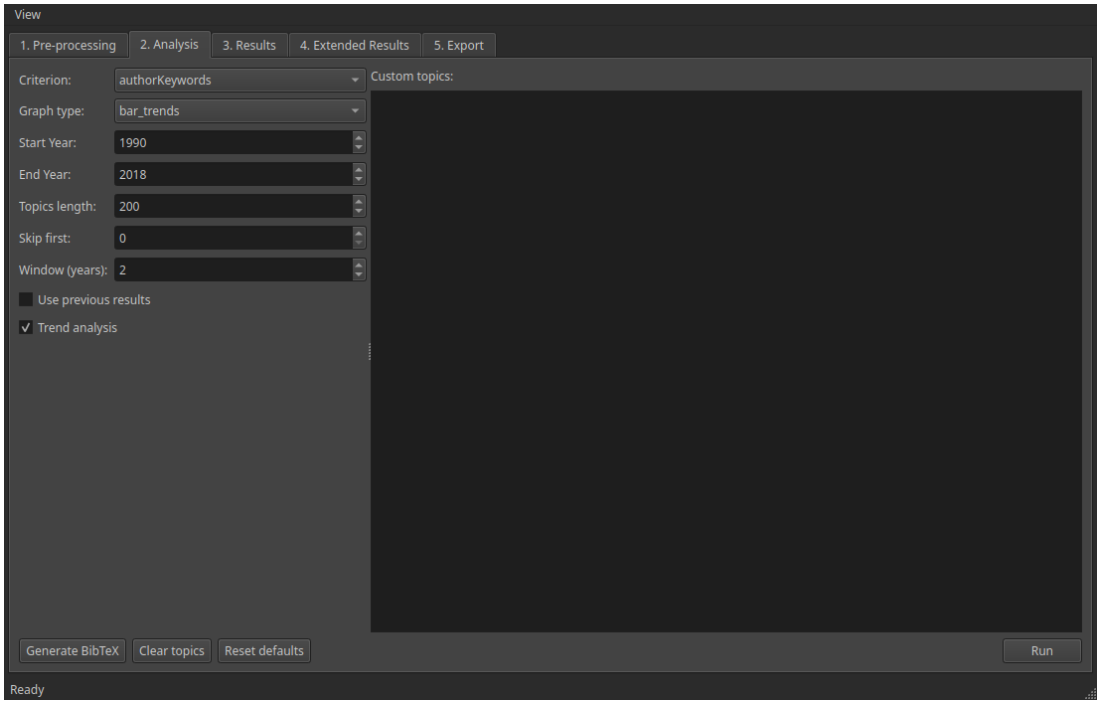
1. Click the `Browse` button, and select the folder that contains the downloaded dataset. The folder is scanned **recursively**, so Scopus and WoS exports can be placed in sub-directories. For the example dataset, select the folder `dataInExample` inside the ScientoPy directory. Alternatively, use the keyboard shortcut **Ctrl+O**.
2. Ensure the `Remove duplicated documents` checkbox is checked (default).
3. Click the `Run preprocess` button (or press **Ctrl+R**). A progress dialog appears and shows each stage of the pipeline: `[1/4] Loading papers`, `[2/4] Disambiguating Scopus author names`, `[3/4] Removing duplicates`, and `[4/4] Saving preprocessed corpus`, with a progress bar per stage.
4. After preprocessing completes, the preprocess brief graph will appear and the application will automatically switch to the **3. Results** tab showing the preprocess brief table.

As result, inside the folder `ScientoPy/dataPre` you will find the following files:

- **papersPreprocessed.parquet:** canonical preprocessed corpus (Apache Parquet, Zstd-compressed). This is the input used by every subsequent analysis. To get a Scopus- or WoS-style CSV copy, use the **5. Export** tab (Section 8).
- **PreprocessedBrief.csv:** this file briefs the pre-process statistics results, such as duplicated papers removed, types of documents, and others.

5 Analysis (Tab 2)

The analysis tab lets you perform scientometric analysis operations. Click on the `2. Analysis` tab to access it.



5.1 Analysis parameters (left panel)

- **Criterion:** the search criterion to analyze, such as author keywords, author names, countries, etc. See Table 1 for the full list.
- **Graph type:** the type of output graph. See Section 11 for details.
- **Start year:** limit the start year for the analysis and graphs.
- **End year:** limit the end year for the analysis and graphs.
- **Topics length:** number of top topics to find in the selected criterion.
- **Skip first:** number of first top topics to skip during the analysis.
- **Window (years):** window width in years for growth indicators (AGR, ADY).
- **Use previous results:** run the analysis based on the output documents from the previous run.
- **Trend analysis:** find the top trending topics with the highest average growth rate.

5.2 Custom topics (right panel)

The right panel contains a text area for entering custom topics. Enter one topic per line. Topics can be grouped with commas and separated by line breaks. The asterisk (*) wildcard is supported. See Section 5.6 for details.

5.3 Action buttons (bottom bar)

- **Generate BibTeX:** generate a BibTeX bibliography file from a LaTeX document (see Section 9).
- **Clear topics:** clear the Custom topics text area.

- **Reset defaults:** reset all analysis parameters to their default values (criterion: authorKeywords, graph type: bar_trends, years: 1990 to last year, topics length: 10, skip first: 0, window: 2, checkboxes unchecked, custom topics cleared).
- **Run:** execute the analysis with the current parameters. Can also be triggered with **Ctrl+R**.

After the analysis completes, results are displayed in the **3. Results** and **4. Extended Results** tabs, and the output graph appears.

5.4 Criterion descriptions

The ScientoPy criterion are described in Table 1:

Table 1: ScientoPy criterion description

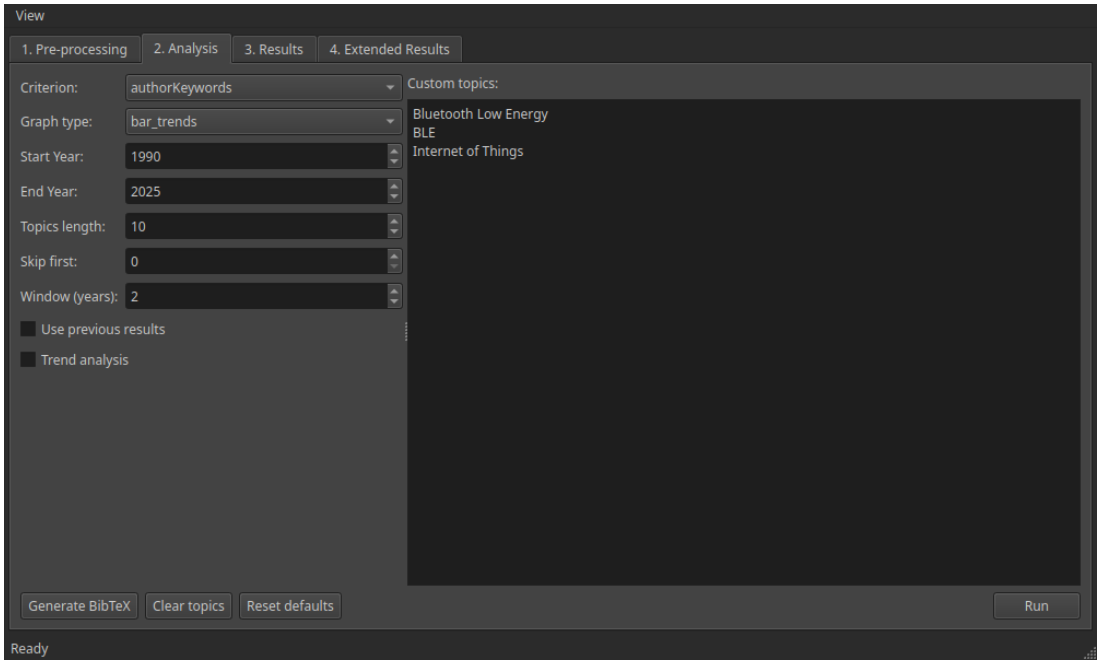
Criterion	Description
author	Authors last name and first name initial
sourceTitle	Publication or journal name
subject	Research areas, only from WoS documents
authorKeywords	Author keywords
indexKeywords	Keywords generated by the index, from WoS *Keyword Plus*, and from Scopus *Indexed keywords*
bothKeywords	AuthorKeywords and indexKeywords are used for this search
abstract	Document abstract, for use with pre-defined topics and asterisk wildcard
documentType	Type of document
dataBase	Database where the document was extracted (WoS or Scopus)
country	Country extracted from authors affiliations
institution	Institution extracted from authors affiliations
institutionWithCountry	Institution with country extracted from authors affiliations

5.5 Extract top topics

To find the top 10 author keywords, keep the default parameters in the analysis tab and click the `Run` button. This will generate a list with the top 10 topics on the selected criterion (in this case authorKeywords), with the number of documents per topic. The bar trends graph will appear, and the quantitative results are saved in the folder `ScientoPy/results` and displayed in the **3. Results** tab.

5.6 Analyze custom topics inside a criterion

If you want to analyze custom topics, such as comparing two countries' publication trends, enter the topics in the `Custom topics` text area (one topic per line):



You can analyze any topic in any criterion. You can also group two or more topics into one by separating them with commas. This is useful for abbreviations and plural/singular forms, for example:

```
WSN, Wireless sensor network, Wireless sensor networks
RFID, RADIO FREQUENCY IDENTIFICATION
```

Tip: You can also use the `Analyze selection` button in the **3. Results** tab to copy selected topic names directly into the Custom topics field (see Section 6).

5.7 Asterisk (*) wildcard

You can use the asterisk wildcard to find phrases or words which start or end with the entered text. For example, entering `device*` in the Custom topics area will match "device", "devices", "device integration", etc.

ScientoPy prints the topics found for the wildcard-based search in the console output. You can copy this information to analyze each specific topic found individually.

5.8 Evolution plot

Select `evolution` as the Graph type to see an evolution plot. This option plots the accumulative documents, average documents per year (ADY), and percentage of documents in the last years (PDLY).

Note: select the right start and end year limits that fit your plot lines best.

5.9 Finding trending topics

ScientoPy can find the top trending topics based on the highest average growth rate (AGR). The AGR is calculated using the following equation:

$$AGR = \frac{\sum_{i=Y_s}^{Y_e} P_i - P_{i-1}}{(Y_e - Y_s) + 1},$$

where:

AGR = Average growth rate;
Y_s = Start year;
Y_e = End year;
P_i = Number of publications on year i.

To find the top trending topics, check the `Trend analysis` checkbox and click `Run`. ScientoPy will find the top 200 topics, calculate the AGR for the specified window period, and sort them from highest to lowest AGR.

5.10 Analysis based on the previous results

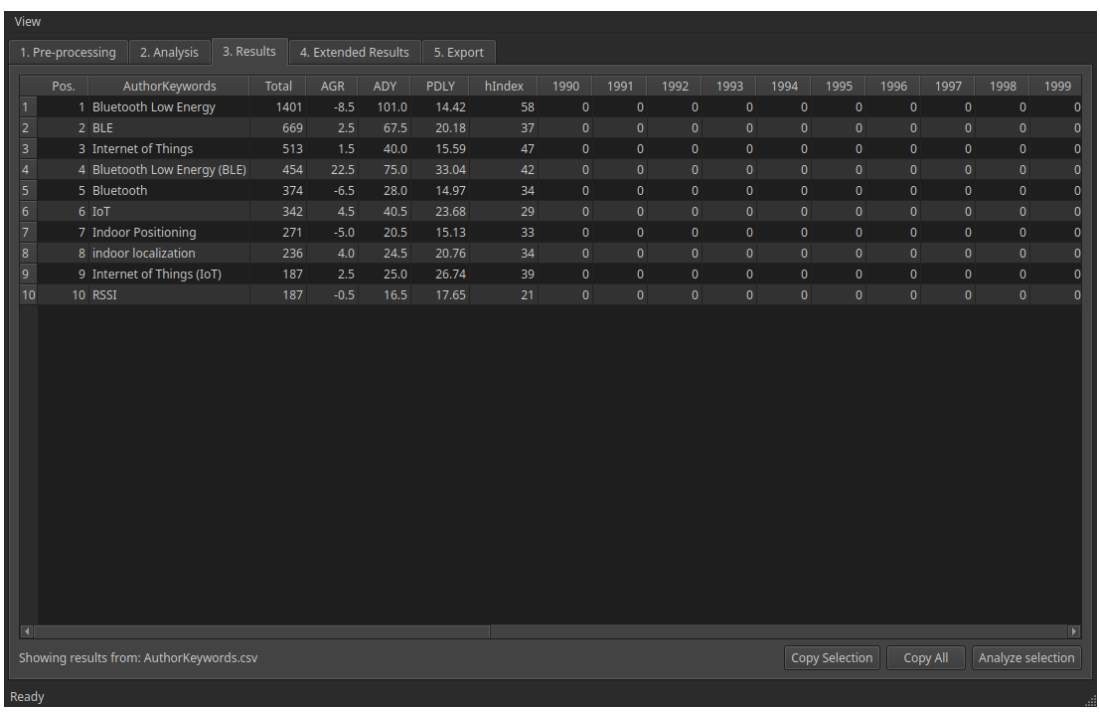
ScientoPy generates a lightweight internal file with all the output documents from the last run so that subsequent analyses can chain off it. For example, if you run an analysis with criterion `country` and custom topic `Canada`, ScientoPy writes `results/lastAnalysis.parquet` with every document whose authors are affiliated in Canada. In previous versions this file was a CSV called `results/papersPreprocessed.csv`; analyses no longer write any CSV automatically. Use the **5. Export** tab (Section 8) if you need a CSV copy.

This output file can be used as input for a subsequent analysis by checking the `Use previous results` checkbox. For example, you could then select criterion `authorKeywords` to find the top author keywords from Canadian papers, or select criterion `country` to find which countries have the most co-authored documents with Canada.

Note: the output documents file is only generated when `Use previous results` is **not** checked. Running multiple analyses with this option checked will not overwrite the documents output file.

6 Results tab (Tab 3)

The Results tab displays the analysis output in a sortable table. After running a preprocess, it shows the preprocessing brief. After running an analysis, it shows the top topics with their statistics (Total, AGR, ADY, PDLY, h-index, and documents per year).



	Pos.	AuthorKeywords	Total	AGR	ADY	PDLY	hIndex	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999
1	1	Bluetooth Low Energy	1401	-8.5	101.0	14.42	58	0	0	0	0	0	0	0	0	0	0
2	2	BLE	669	2.5	67.5	20.18	37	0	0	0	0	0	0	0	0	0	0
3	3	Internet of Things	513	1.5	40.0	15.59	47	0	0	0	0	0	0	0	0	0	0
4	4	Bluetooth Low Energy (BLE)	454	22.5	75.0	33.04	42	0	0	0	0	0	0	0	0	0	0
5	5	Bluetooth	374	-6.5	28.0	14.97	34	0	0	0	0	0	0	0	0	0	0
6	6	IoT	342	4.5	40.5	23.68	29	0	0	0	0	0	0	0	0	0	0
7	7	Indoor Positioning	271	-5.0	20.5	15.13	33	0	0	0	0	0	0	0	0	0	0
8	8	indoor localization	236	4.0	24.5	20.76	34	0	0	0	0	0	0	0	0	0	0
9	9	Internet of Things (IoT)	187	2.5	25.0	26.74	39	0	0	0	0	0	0	0	0	0	0
10	10	RSSI	187	-0.5	16.5	17.65	21	0	0	0	0	0	0	0	0	0	0

6.1 Column tooltips

Hover over column headers to see tooltips with definitions of abbreviations:

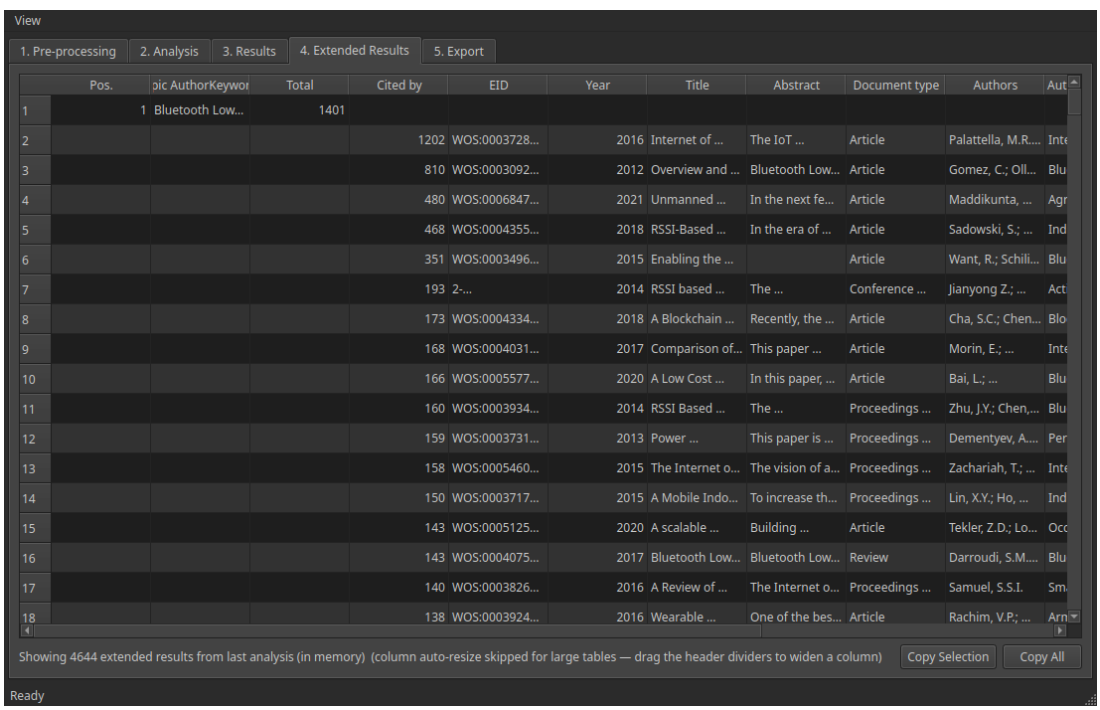
- **AGR:** Average Growth Rate – mean year-over-year change in publications within the window period.
- **ADY:** Average Documents per Year – mean publications per year within the window period.
- **PDLY:** Percentage of Documents in Last Years – proportion of total documents published within the window period.
- **hIndex:** h-index – the largest number h such that h documents have at least h citations each.

6.2 Action buttons

- **Copy Selection:** copy the selected cells to the clipboard (also available via Ctrl+C).
- **Copy All:** copy the entire table to the clipboard.
- **Analyze selection:** copy the topic names from the selected rows to the `Custom topics` text area in the Analysis tab, and switch to the Analysis tab. This is useful for quickly selecting topics from the results for further analysis with a different graph type or criterion.

7 Extended Results tab (Tab 4)

The Extended Results tab shows the detailed per-document output after running an analysis. Each row represents a paper associated with one of the analyzed topics, including authors, title, year, source, citations, DOI, and other metadata.



	Pos.	pic AuthorKeywor	Total	Cited by	EID	Year	Title	Abstract	Document type	Authors	Aut
1	1	Bluetooth Low...	1401								
2				1202	WOS:0003728...	2016	Internet of ...	The IoT ...	Article	Palattella, M.R....	Inte
3				810	WOS:0003092...	2012	Overview and ...	Bluetooth Low...	Article	Gomez, C.; Oll...	Blu
4				480	WOS:0006847...	2021	Unmanned ...	In the next fe...	Article	Maddikunta, ...	Agr
5				468	WOS:0004355...	2018	RSSI-Based ...	In the era of ...	Article	Sadowski, S.; ...	Ind
6				351	WOS:0003496...	2015	Enabling the ...		Article	Want, R.; Schili...	Blu
7				193	2-...	2014	RSSI based ...	The ...	Conference ...	Jianyong Z.; ...	Act
8				173	WOS:0004334...	2018	A Blockchain ...	Recently, the ...	Article	Cha, S.C.; Chen...	Blo
9				168	WOS:0004031...	2017	Comparison of...	This paper ...	Article	Morin, E.; ...	Inte
10				166	WOS:0005577...	2020	A Low Cost ...	In this paper, ...	Article	Bai, L.; ...	Blu
11				160	WOS:0003934...	2014	RSSI Based ...	The ...	Proceedings ...	Zhu, J.Y.; Chen,...	Blu
12				159	WOS:0003731...	2013	Power ...	This paper is ...	Proceedings ...	Dementyev, A....	Per
13				158	WOS:0005460...	2015	The Internet o...	The vision of a...	Proceedings ...	Zachariah, T.; ...	Inte
14				150	WOS:0003717...	2015	A Mobile Indo...	To increase th...	Proceedings ...	Lin, X.Y.; Ho, ...	Ind
15				143	WOS:0005125...	2020	A scalable ...	Building ...	Article	Tekler, Z.D.; Lo...	Occ
16				143	WOS:0004075...	2017	Bluetooth Low...	Bluetooth Low...	Review	Darroudi, S.M....	Blu
17				140	WOS:0003826...	2016	A Review of ...	The Internet o...	Proceedings ...	Samuel, S.S.I.	Sm
18				138	WOS:0003924...	2016	Wearable ...	One of the bes...	Article	Rachim, V.P.; ...	Arn

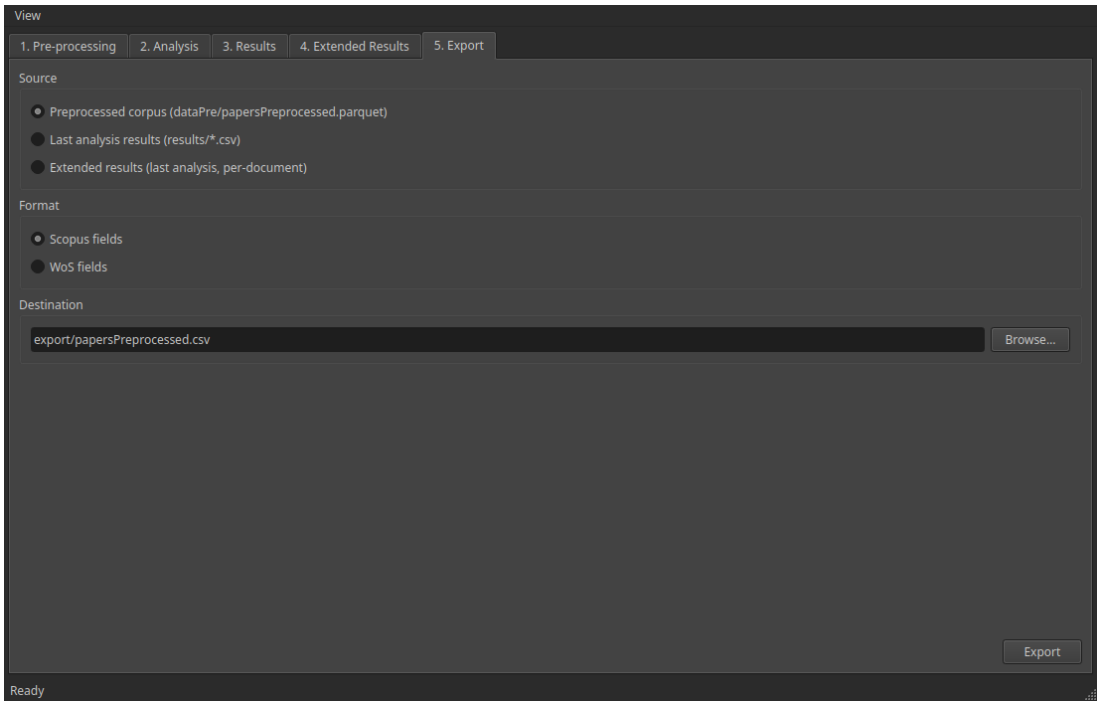
The table is sortable by clicking column headers and supports cell selection with Ctrl+C copy.

7.1 Action buttons

- **Copy Selection:** copy the selected cells to the clipboard.
- **Copy All:** copy the entire table to the clipboard.

8 Export tab (Tab 5)

The Export tab produces on-demand Scopus- or WoS-style CSV files from either the full preprocessed corpus or the most recent analysis results. Exports are opt-in: neither preprocessing nor analysis writes CSV by itself anymore (the canonical on-disk format is Parquet, which is much smaller and loads faster).



1. **Source.**
   - **Preprocessed corpus** – export `dataPre/papersPreprocessed.parquet` as a single CSV.
   - **Last analysis results** – copy all CSVs currently under `results/` into the chosen destination folder.
   - **Extended results (last analysis, per-document)** – write the detailed per-document `*_extended.csv` for the analysis that is still in memory from the last `Run`. This file is no longer produced automatically because it can reach hundreds of thousands of rows on large corpora. The option is disabled until you have run an analysis in the current session.
2. **Format.** Only applies to *Preprocessed corpus*; greyed out for the other sources.
   - **Scopus fields** – Scopus column-naming convention (`Authors`, `Title`, `EID`, ...).
   - **WoS fields** – Web of Science field tags (`AU`, `TI`, `UT`, ...).
3. **Destination.** A file path for Preprocessed corpus (default `export/papersPreprocessed.csv`) or Extended results (default `export/<Criterion>_extended.csv`), or a folder path for Last analysis results (default `export/results/`). Use the `Browse...` button to pick an alternative path.
4. **Export.** Click the `Export` button. A progress dialog appears and updates through the stages `Reading preprocessed corpus...` and `Writing CSV (scopus fields)` with a live 0%–100% bar during the write. On completion, a confirmation dialog offers an `Open` button to reveal the output file or folder.

The `Export` button is automatically disabled when the chosen Source has nothing to export (e.g. no preprocessed corpus has been produced yet, or `results/` is empty). The same export functionality is available from the command line via `exportPapers.py` (run `python3 exportPapers.py -h` for the full list of options).

9 Generate BibTeX

With ScientoPy you can generate a LaTeX BibTeX file for your references. Follow the next steps:

1. Run an analysis to generate results.
2. Open the Extended Results tab (Tab 4) and find the EID values of the papers you want to cite.
3. In your LaTeX document, use the paper EID values as citation keys inside `\cite{}` commands, for example: `\cite{2-s2.0-85012345678}`
4. Save your LaTeX document.
5. In the Analysis tab, click the `Generate BibTeX` button and select the LaTeX document.
6. ScientoPy will find all EID citation keys in the document and generate a BibTeX file with the corresponding bibliographic entries. The output file is saved next to the input file as `<filename>_bibliography.bib`.

Note: for your reference we have included an article LaTeX file example in `latexExample/article_example.tex`. This contains cites from the example dataset. To test the Generate BibTeX capability:

- Preprocess the example dataset located in `dataInExample`
- In the Analysis tab, click `Generate BibTeX`
- Select the file `latexExample/article_example.tex`
- Compile the LaTeX file with: `latexmk -pdf article_example.tex`

10 Output files and directories

After running ScientoPy analysis you will find the following folder and files structure:

```
ScientoPy
├── dataInExample
├── dataPre
│   ├── papersPreprocessed.csv
│   └── PreprocessedBrief.csv
├── graphs
├── latexExample
├── Manual
├── results
│   ├── AuthorKeywords.csv
│   ├── AuthorKeywords_extended.csv
│   └── papersPreprocessed.csv
└── scientopy_config.json
```

These folders and output files are described below:

- **dataInExample:** contains Scopus and WoS example data set for the search criteria "Bluetooth low energy" downloaded on April 2, 2026. This is the input example for preprocessing.
- **dataPre:** output folder for the preprocessing results, and input folder for the analysis.
- **papersPreprocessed.csv:** preprocessed papers data with all input documents merged, filtered, and duplicates removed. This is the input file that the analysis uses.
- **PreprocessedBrief.csv:** preprocessing brief table that shows the preprocessing results related to total papers found per database, the omitted papers, the duplicated papers count per database, and the total number of papers per paper type (Conference paper, article, review...).
- **graphs:** graphs output folder for preprocessing and analysis.
- **latexExample:** folder with the LaTeX file article example to generate the BibTeX from the preprocessed dataset.
- **Manual:** folder with the PDF manual.
- **results:** output folder for analysis result files.
- **AuthorKeywords.csv:** analysis output file for the selected criterion (in this case authorKeywords) that shows the top or custom topics with the total number of documents, the Average Growth Rate (AGR), the Average Documents per Year (ADY), the Percentage of Documents in Last Years (PDLY), the h-index, and the documents per each year.
- **AuthorKeywords_extended.csv:** analysis output file for the selected criterion that shows the top or custom topics with the documents related to each one.
- **papersPreprocessed.csv:** inside the results folder, this file contains the output papers from the last analysis run. This is used as input when the option `Use previous results` is checked.
- **scientopy_config.json:** user configuration file storing window geometry, appearance mode, and analysis parameters. Automatically created and updated.

11 ScientoPy graph types

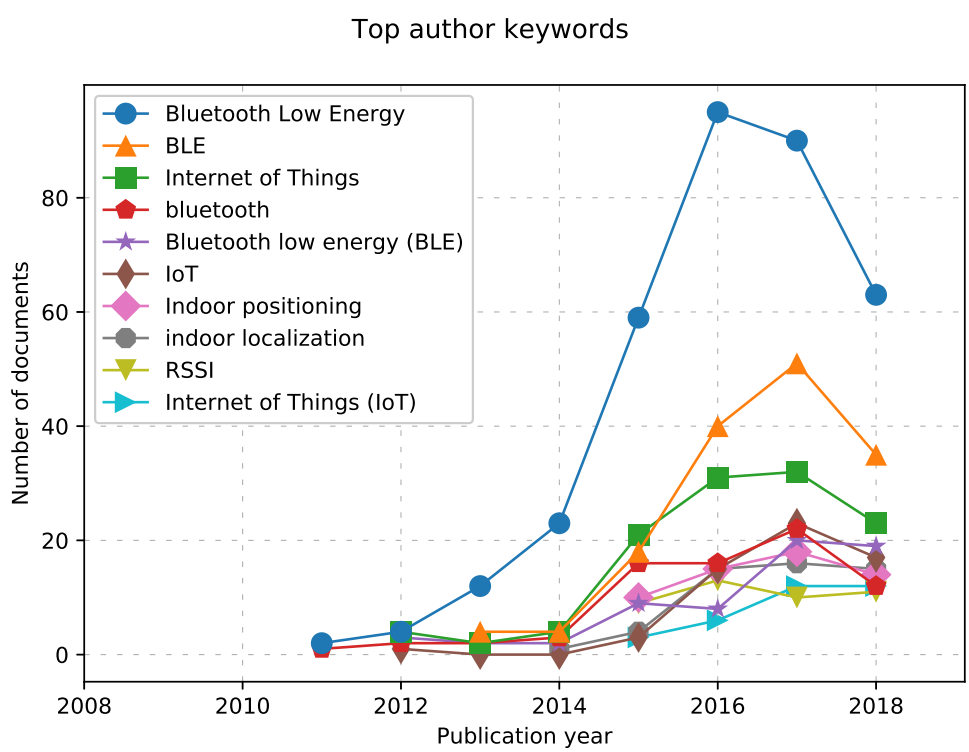
ScientoPy has 5 different ways to graph the results described on Table 2.

Table 2: ScientoPy output graph types

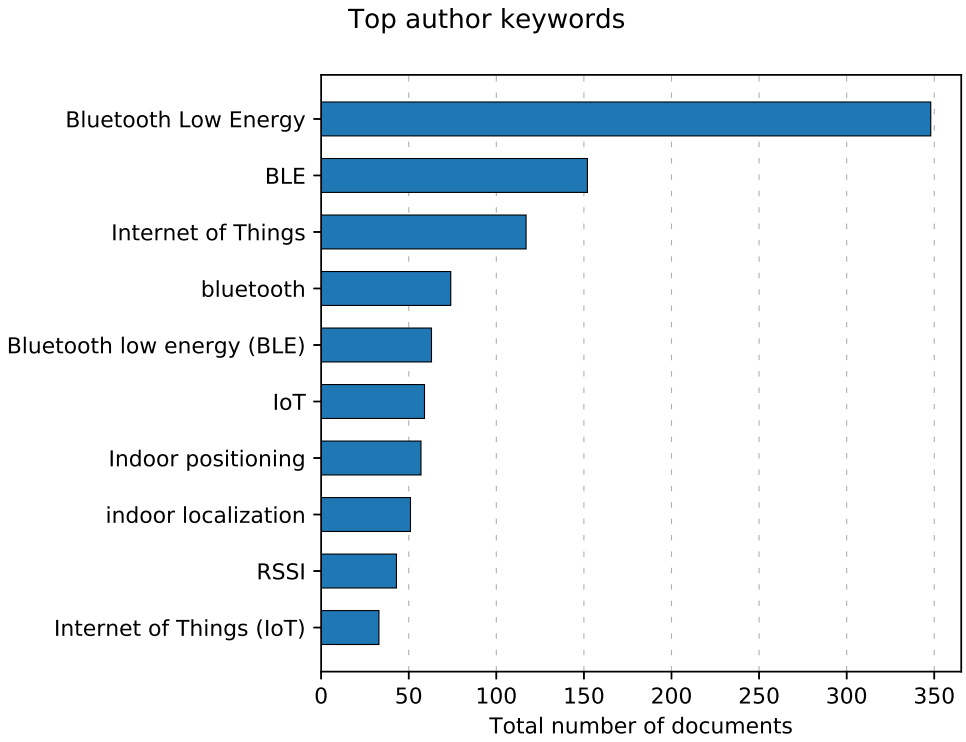
Graph type	Abbreviation	Description
Time line	`time_line`	Graphs the number of documents of each topic vs the publication year
Horizontal bars	`bar`	Graphs the total number of documents of each topic in horizontal bars
Horizontal bars trends	`bar_trends`	Graphs the total number of documents of each topic in horizontal bars, with the percentage of documents published in the last years
Evolution	`evolution`	Graphs two plots, one with the accumulative number of documents vs the publication year, and another with the average papers per year vs the percentage of documents in the last years
Word cloud	`word_cloud`	Generate a word cloud based on the topic total number of publications

Below are some examples of these graph types.

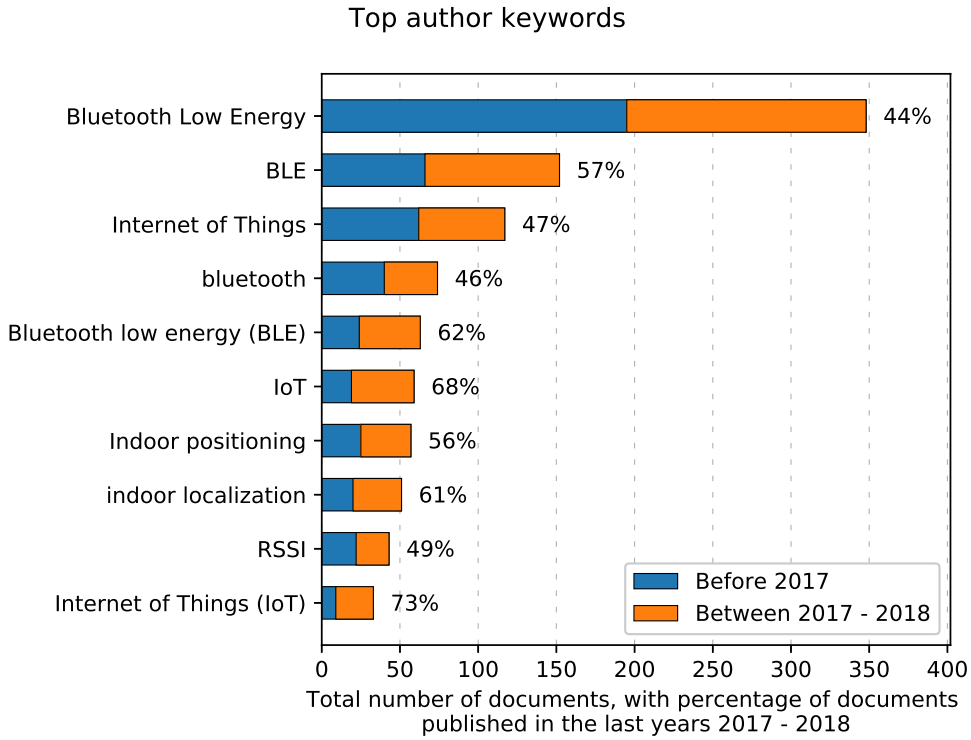
11.1 Time line graph



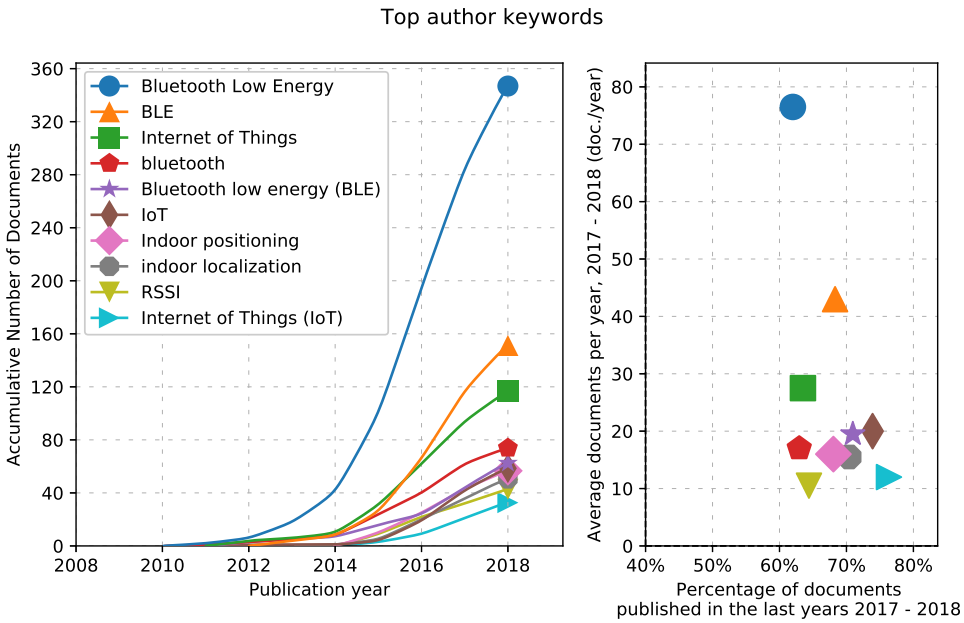
11.2 Horizontal bars graph



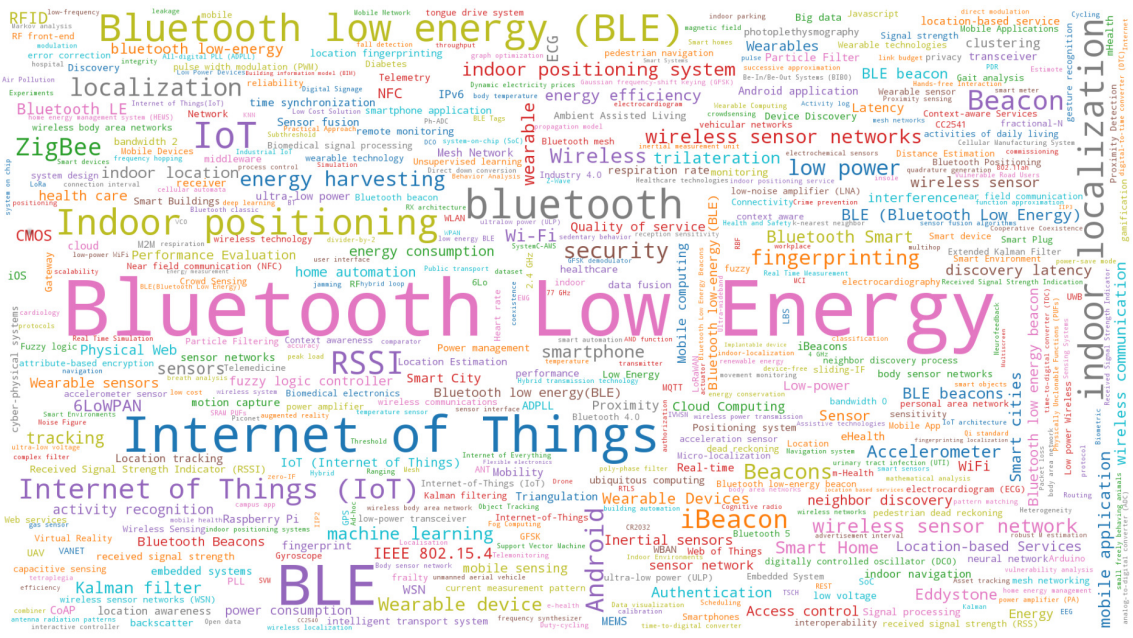
11.3 Horizontal bars trends



11.4 Evolution graph



11.5 Word cloud graph


Bluetooth low energy (BLE)
Bluetooth Low Energy
Internet of Things
Internet of Things (IoT)
localization
IoT
ZigBee
Indoor positioning
bluetooth
energy harvesting
wireless sensor networks
Wireless trilateration
low power
Beacon
indoor localization
security
RSSI
BLE
iBeacon
wireless sensor network
machine learning
Wearable device
Android
Wearables
clustering
indoor positioning system
energy efficiency
fingerprinting
discovery latency
smartphone
Accelerometer
Smart Home
Location-based Services
Authentication
Eddystone
Access control
Kalman filter
Bluetooth Beacons
IEEE 802.15.4
mobile sensing
Wearable Devices
neighbor discovery
Bluetooth Smart
BLE (Bluetooth Low Energy)
Wi-Fi
Performance Evaluation
home automation
energy consumption
Physical Web
sensor networks
Wearable sensors
6LoWPAN
tracking
activity recognition
Raspberry Pi
embedded systems
CoAP
location awareness
power consumption
backscatter
intelligent transport system
Smart Cities
mobile application
wireless communication
Bluetooth low energy beacon
Smart Environment
Low-power
BLE beacons
Cloud Computing
Sensor
Positioning system
Real-time
Beacons
WiFi
Inertial sensors
Web of Things
digitally controlled oscillator (DCO)
sensor network
Embedded System
indoor navigation
Smartphones
Energy
interoperability
received signal strength (RSS)
Signal processing
Time synchronization
Smartphone application
Sensor fusion
remote monitoring
Mesh Network
wearable technology
Unsupervised learning
Biomedical signal processing
Ambient Assisted Living
Bluetooth mesh
Distance Estimation
Context-aware Services
Bluetooth Positioning
Cellular Manufacturing System
Mobile Devices
middleware
receiver
health care
Smart Buildings
ultra-low power
Bluetooth beacon
CMOS
cloud
Near field communication (NFC)
Crowd Sensing
Gateway
iOS
Fuzzy logic
Particle Filtering
Context awareness
Location Estimation
Power management
fuzzy logic controller
Smart City
performance
Low Energy
motion capture
Proximity
Bluetooth low energy(BLE)
Bluetooth 4.0
Location tracking
IoT (Internet of Things)
Internet-of-Things (IoT)
Triangulation
Kalman filtering
Mobility
ubiquitous computing
Raspberry Pi
machine learning
Gyroscope
fingerprint
mobile sensing
WSN
current measurement pattern
Wireless sensing
Virtual Reality
UAV
VANET
received signal strength
capacitive sensing
efficiency
PLL
SVM
wireless sensor networks (WSN)
antenna radiation patterns
interactive controller
Open data
wireless localization
Frequency synthesizer
Duty-cycling
MEMS
time-to-digital converter
Smart Plug
Extended Kalman Filter
Smart device
Cooperative Coexistence
Received Signal Strength Indication
Proximity Detection
gamification
digital-to-time converter (DTC)
Internet
mHealth
gesture recognition
Cycling
Mobile Applications
Signal strength
photoplethysmography
Wearable technologies
Javascript
Big data
location-based service
Gait analysis
BLE beacon
transceiver
Particle Filter
pedestrian navigation
Telemetry
NFC
IPv6
Diabetes
Digital Signage
reliability
Network
Internet of Things(IoT)
Wireless body area networks
Bluetooth LE
bandwidth 2
Mobile Devices
frequency hopping
Industrial IoT
Subthreshold
Practical Approach
Biomedical signal processing
Mesh Network
Unsupervised learning
Industry 4.0
Z-Wave
Bluetooth mesh
Wireless trilateration
respiration rate
Healthcare technologies
indoor positioning service
low-noise amplifier (LNA)
Connectivity
Crime prevention
context aware
Health and Safety
k-nearest neighbor
Bluetooth Smart
Smart device
Smart Plug
Extended Kalman Filter
Smart Environment
Real Time Measurement
electrocardiography
Received Signal Strength Indication
discovery latency
healthcare
data fusion
smartphone
Mobile computing
Bluetooth low energy(BLE)
Low-power
BLE beacons
Cloud Computing
Sensor
Positioning system
Real-time
Beacons
WiFi
Accelerometer
Smart Cities
Bluetooth low energy beacon
Wireless communication
mobile application